

Position: Current Model Licensing Practices are Dragging Us into a Quagmire of Legal Noncompliance

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Background & Problem — We show reuse is exploding and licensing lags

- We highlight rapid model reuse (pretraining, PEFT, merges) powering fast customization.
- We argue licensing is foundational to AI supply-chain risk and governance.
- We motivate that unstandardized licenses create cascading downstream risk.

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Why This Is Hard — We trace obligations through transitive graphs

- We explain how merges and multi-hop reuse propagate obligations.
- We show implicit dependencies obscure provenance and compliance.
- We visualize the global graph (left), merge edges (middle), and a 3-hop Llama-3.2-3B-Instruct neighborhood (right).



Figure: Visualization of Model Dependencies on HF — fig1

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Our Position & Contributions — We audit, we quantify, we propose

- We take the position that current practices pull ML into a noncompliance quagmire.
- We contribute a large-scale audit and a rule-based detection pipeline.
- We propose a clarity framework and MGLs + Model Sheets to reduce conflicts.

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Methodology & Rules — We encode reuse semantics and conflict families

- We analyze a Dec 2024 Hugging Face snapshot and build a reuse graph.
- We operationalize reuse—Finetune, Adapter, Quantization, Merge—and treat finetune/quantization/merge as derivatives; adapters as add-ons.
- We check conflicts across families: ND, FSFf, CCf, and Llama exclusivity.

Key Quant Findings — We find mismatch dominates; conflicts concentrate

- We find 81.8% license mismatch (dominant issue).
- Copyleft-style term issues are frequent where applicable (63.5%).
- We find Llama exclusivity hotspots concentrate around merges and relabeling.

Table: Table 3. Quantitative Results of Noncompliance. NCM: NonCompliance Models, CTM: Corrected Total Models.

Noncompliance	# of NCM	# of CTM	Finetune	Merge	Quantization	Adapter
License Mismatch	123,707 (81.8%)	151,302	n/a	n/a	n/a	n/a
Copyleft-style Terms	26,679 (63.5%)	41,019	11,011 (35.8%)	1,453 (4.7%)	4,775 (15.5%)	13,546 (44.0%)
Copyleft Violation	365 (38.5%)	949	296 (81.1%)	32 (8.8%)	37 (10.1%)	n/a
ND	36 (100%)	36	1 (2.8%)	12 (33.3%)	23 (63.9%)	n/a
CCf	76 (2.1%)	3,675	17 (20.0%)	67 (78.8%)	1 (1.2%)	n/a
FSFf	98 (45.8%)	214	84 (85.7%)	0	14 (14.3%)	n/a
La2E-license	1,000 (38.4%)	2,602	182 (18.1%)	29 (2.9%)	331 (32.9%)	465 (46.2%)
La2E-merge	11 (55.0%)	20	n/a	11 (100%)	n/a	n/a
La3E-license	966 (24.7%)	3,904	518 (47.2%)	83 (7.6%)	303 (27.6%)	193 (17.6%)
La3E-merge	81 (56.6%)	143	n/a	81 (100%)	n/a	n/a

Conflict Families & Spread — We show how conflicts propagate

- We find merges concentrate and propagate conflicts across trees.
- We show ND is absolute once triggered.
- We observe FSFf and Llama exclusivity dominate detected conflicts.



Figure: Models with License Conflicts and Their Derivatives.

Case Study: Llama-3.2-3B-Instruct — We see license switches and nulls

- We observe license switches where behavioral clauses get lost downstream.
- We see many null licenses, creating uncertainty and breach risks.
- We illustrate how choices compound in second-order derivatives.



Figure: License distribution among direct derivatives of Llama-3.2-3B-Instruct.

- How should we scope derivative for merges, adapters, and quantization?
- What is a practical boundary for RAI/acceptable-use terms without breaking GPL freedoms?
- How do we track implicit dependencies and knowledge transfer at scale?
- What minimal metadata & AI BOM fields enable auditable compliance?
- How should hubs surface or enforce conflicts to developers?